WEB TECHNOLOGY AND MOBILE APPLCATION

EXPERIMENT 9

Calculator App

Aim:

To develop an Android application using Button, EditText, and TextView controls to perform basic

arithmetic operations such as:

- **Addition**
- **Subtraction**
- **Multiplication**
- **Division**

CREATE CALCULATOR PROJECT

Step 7: Open Android Studio

Click:

New Project

Step 8: Choose Template

Select:

Empty Views Activity

Click Next

Step 9: Fill Project Details

Fill exactly like this:

Field	Value
Name	CalculatorApp

Package Name **com.example.calculatorapp**

Language **Java**

Minimum SDK **API 24 or change API 21**

Click:

Finish

Wait until bottom shows:

Build Successful

PHASE 4 – UNDERSTAND PROJECT STRUCTURE

On left side:

Change view to:

Android

You will see:

app

```
├── java
│ └── MainActivity.java
├── res
│ └── layout
│ └── activity_main.xml
```

PHASE 5 – DESIGN USER INTERFACE (XML)

Step 10: Open Layout File

Go to:

res → layout → activity_main.xml

Click:

Code (top right)

Delete existing code.

Paste this:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
android:padding="20dp"
android:layout_width="match_parent"
android:layout_height="match_parent">

<EditText
android:id="@+id/num1"
android:hint="Enter First Number"
android:inputType="numberDecimal"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

<EditText
android:id="@+id/num2"

android:hint="Enter Second Number"
android:inputType="numberDecimal"
android:layout_width="match_parent"
android:layout_height="wrap_content"
```

```
android:layout_marginTop="10dp"/>

<TextView
android:id="@+id/result"
android:text="Result"
android:textSize="20sp"
android:layout_marginTop="20dp"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

<Button
android:text="Add"
android:onClick="add"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

<Button
android:text="Subtract"
android:onClick="subtract"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

<Button
android:text="Multiply"
android:onClick="multiply"
android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"/>

<Button
android:text="Divide"
android:onClick="divide"
android:layout_width="match_parent"
android:layout_height="wrap_content"/>

</LinearLayout>
```

Press:

Ctrl + S

PHASE 6 – WRITE JAVA LOGIC

Step 11: Open Java File

Go to:

java → com.example.calculatorapp → MainActivity.java

Delete everything.

Paste this:

```
package com.example.calculatorapp;

import android.os.Bundle;
import android.view.View;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

EditText num1, num2;
TextView result;

@Override
protected void onCreate(Bundle savedInstanceState) {
super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

num1 = findViewById(R.id.num1);
num2 = findViewById(R.id.num2);
result = findViewById(R.id.result);
}

public void add(View v) {
double a = Double.parseDouble(num1.getText().toString());
double b = Double.parseDouble(num2.getText().toString());
result.setText("Result: " + (a + b));
}

public void subtract(View v) {
double a = Double.parseDouble(num1.getText().toString());
double b = Double.parseDouble(num2.getText().toString());
result.setText("Result: " + (a - b));
```

```
}

public void multiply(View v) {
double a = Double.parseDouble(num1.getText().toString());
double b = Double.parseDouble(num2.getText().toString());
result.setText("Result: " + (a * b));
}

public void divide(View v) {
double a = Double.parseDouble(num1.getText().toString());
double b = Double.parseDouble(num2.getText().toString());

if (b == 0) {
result.setText("Cannot divide by zero");
} else {
result.setText("Result: " + (a / b));
}
}
}
```

Press:

Ctrl + S

PHASE 7 – RUN THE APPLICATION

Step 12: Create Emulator

Click:

Device Manager

Create Device

Choose Pixel 4

Next

Download system image

Finish

Step 13: Run Application

Click:

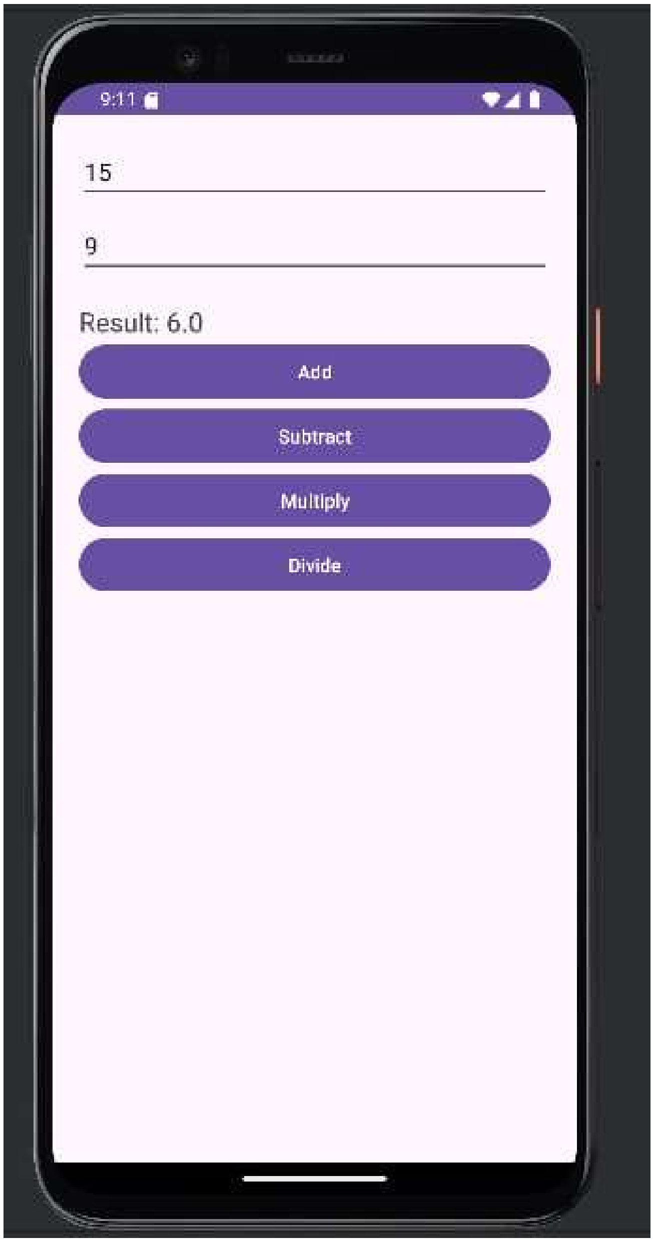
Green ▶ Run button

Wait for emulator to open.

OUTPUT

Application displays:

- **Two input fields**
- **Four buttons**
- **Result display**


9:11
15
9
Result: 6.0
Add
Subtract
Multiply
Divide